

Sequence Types

COMS 261

In Python, lists, strings, and ranges are all examples of **sequence types**. Sequence types store items in a specific order, where each item can be accessed by its position (**index**).

1. How would you print the 2nd word in `my_list`?

```
| my_list = ["This", "is", "a", "list"]
```

A list is **mutable** which means that you can change its elements without creating a whole new list. You can't do that with strings because they are **immutable**.

2. What will the following commands do? Use a computer to test your prediction.

```
| my_list = ["This", "is", "a", "list"]
| my_list[2] = "my"
| print(my_list)
```

3. What will the following commands do? Use a computer to test your prediction.

```
| my_string = "This is a string."
| my_string[2] = "!"
| print(my_string)
```

To add an element to a list, you can use the `append` method or the `+` operator. Assuming that `t` is a list and `x` is an element, both of these are correct:

```
| t.append(x)          # Appending mutates (changes) an existing list
| t = t + [x]          # Adding two lists together creates a new list
```

But these are wrong:

```
| t.append([x])        # WRONG!
| t = t.append(x)      # WRONG!
| t + [x]              # WRONG!
| t = t + x            # WRONG!
```

4. Create a list `t` and an element `x` to test the four wrong commands above.
 - (a) Which command gives you an error message? Why?

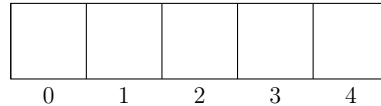
- (b) What do the other three wrong commands do? Why don't they work?

The **range** function creates a **range type** object. Range types aren't lists, but they are sequences. They are also immutable.

5. What happens when you run these commands?

```
my_range = range(1, 4)
print(my_range[2])
```

6. What are the elements of **range(5, 10)**? Put the elements into the diagram below.



You can convert any sequence type into a list using the **list** function.

7. What are the values of the following Python expressions? Use the shell to check your answers.

(a) **list("abc")**

(b) **list(range(4))**

(c) **list(5)**